



GATE ME 90 Days Challenge

Day 4/90 · Practice Set with Detailed Solutions

SUBJECT

Engineering Mechanics

TODAY'S TOPIC

Impulse, Energy & Impact

COUNTDOWN

107 days to GATE

How to use this set: 15 questions (7 MCQ, 8 NAT) on impulse–momentum (linear and angular), energy methods, collisions and virtual work. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's free test: 7:00 PM on our website and app, 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

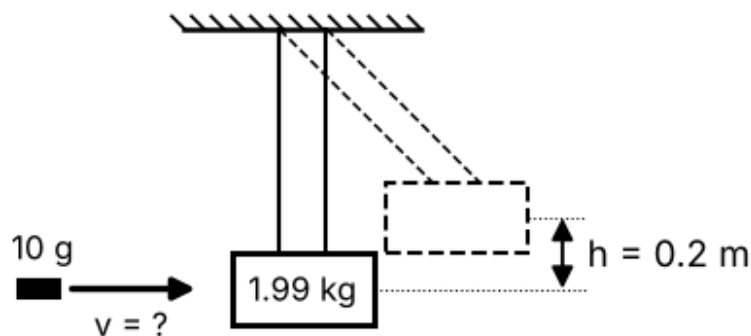
Questions

Q1 (NAT)

A 20 g bullet moving horizontally at 400 m/s embeds itself in a 2 kg block resting on a smooth horizontal floor. The common velocity just after impact (in m/s) is _____

Q2 (MCQ)

A 10 g bullet is fired into a 1.99 kg wooden block hanging from light strings (ballistic pendulum) and stays inside it. The block rises by 0.2 m, as shown. The speed of the bullet was ($g = 9.81 \text{ m/s}^2$)



(A) 198.1 m/s

(B) 396.2 m/s

(C) 280.1 m/s

(D) 39.6 m/s

Q3 (NAT)

A ball dropped from a height of 4 m on to a hard floor rebounds to a height of 2.25 m. The coefficient of restitution between the ball and the floor is _____

Q4 (MCQ)

A 2 kg sphere moving at 6 m/s makes a direct central impact with a 4 kg sphere at rest. If $e = 0.5$, the velocities of the 2 kg and 4 kg spheres after impact are

(A) 2 m/s and 2 m/s

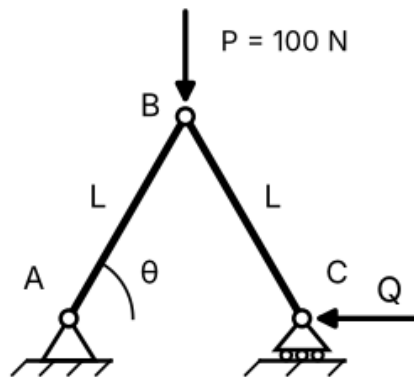
(B) -2 m/s and 4 m/s

(C) 1 m/s and 2.5 m/s

(D) 0 m/s and 3 m/s

Q10 (MCQ)

Two equal light links AB and BC are pinned together at B. A is a pin and C is a roller on the same level, as shown. A vertical load $P = 100 \text{ N}$ acts at B. Using the principle of virtual work, the horizontal force Q at C needed for equilibrium when $\theta = 60^\circ$ is

**(A)** 57.7 N**(B)** 86.6 N**(C)** 28.9 N**(D)** 50.0 N**Q11** (NAT)

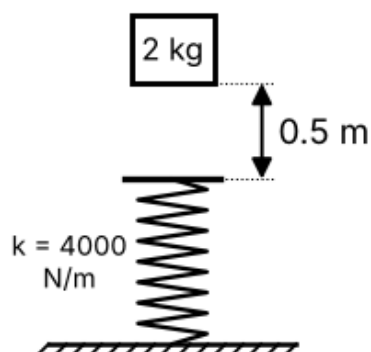
A ball strikes a smooth horizontal floor at 10 m/s , its path making 30° with the floor. If $e = 0.6$, the speed of the ball just after the bounce (in m/s) is _____

Q12 (MCQ)

A uniform rod of mass 3 kg and length 1 m lies at rest on a smooth horizontal table. A horizontal impulse of $6 \text{ N}\cdot\text{s}$ is applied at one end, perpendicular to the rod. The angular velocity of the rod just after the blow is

(A) 6 rad/s **(B)** 12 rad/s **(C)** 4 rad/s **(D)** 24 rad/s **Q13** (NAT)

A 2 kg block is dropped from rest 0.5 m above the top of a vertical spring of stiffness 4000 N/m , as shown. The maximum compression of the spring (in mm) is _____



Q14 MCQ

A smooth sphere moving on a smooth table strikes an identical sphere at rest in a glancing (oblique) **perfectly elastic** collision. After impact the two spheres move

- (A) along the same line
(B) in directions at 90° to each other
(C) in directions at 45° to each other
(D) with equal speeds always

Q15 NAT

A 20 kg block rests on a horizontal floor ($\mu_s = \mu_k = 0.2$). A horizontal force $P = 10t$ newtons (t in s) is applied from $t = 0$. The speed of the block at $t = 8$ s (in m/s) is _____

Answer Key

Q	Type	Answer
Q1	NAT	3.96 (accept 3.94 to 3.98)
Q2	MCQ	(B) 396.2 m/s
Q3	NAT	0.75
Q4	MCQ	(D) 0 m/s and 3 m/s
Q5	NAT	50
Q6	MCQ	(C) 94.9 mm
Q7	NAT	5.06 (accept 5.02 to 5.10)
Q8	MCQ	(C) 6.25 rad/s
Q9	NAT	9.905 (accept 9.85 to 9.95)
Q10	MCQ	(C) 28.9 N
Q11	NAT	9.165 (accept 9.10 to 9.22)
Q12	MCQ	(B) 12 rad/s
Q13	NAT	75.0 (accept 74 to 76)
Q14	MCQ	(B) in directions at 90° to each other
Q15	NAT	4.15 (accept 4.10 to 4.20)

Detailed Solutions

Q1 Answer: 3.96 (accept 3.94 to 3.98)

Concept: During the very short impact, external horizontal impulse is zero, so linear momentum is conserved (energy is not).

$$0.02 \times 400 = (0.02 + 2) v$$

$$v = 8/2.02 = \mathbf{3.96 \text{ m/s}}$$

KE before = $\frac{1}{2}(0.02)(400^2) = 1600 \text{ J}$; KE after = $\frac{1}{2}(2.02)(3.96^2) = 15.8 \text{ J} \rightarrow$ about 99 % is lost as heat and deformation.

Exam tip: In a plastic impact with a heavy target, the fraction of KE lost is $M/(m + M)$, nearly all of it.

Q2 Answer: (B) 396.2 m/s

Concept: Two stages. Impact: momentum conserved. Swing: mechanical energy conserved.

Swing: $V = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 0.2} = \mathbf{1.981 \text{ m/s}}$ (block + bullet just after impact)

Impact: $0.01 u = (0.01 + 1.99) V \rightarrow u = 200 \times 1.981 = \mathbf{396.2 \text{ m/s}}$

Exam tip: Never apply energy conservation across the impact itself; KE is lost when the bullet embeds.

Q3 Answer: 0.75

Concept: $e = (\text{speed of separation})/(\text{speed of approach})$. For a fixed floor $e = v/u$.

$$u = \sqrt{2g \times 4}, v = \sqrt{2g \times 2.25}$$

$$e = \sqrt{2.25/4} = 1.5/2 = \mathbf{0.75}$$

Exam tip: $e = \sqrt{h_2/h_1}$, not h_2/h_1 (0.5625 is the trap).

Q4 Answer: (D) 0 m/s and 3 m/s

Concept: Use momentum conservation and Newton's law of restitution together.

$$\text{Momentum: } 2(6) = 2v_1 + 4v_2 \rightarrow v_1 + 2v_2 = 6$$

$$\text{Restitution: } v_2 - v_1 = 0.5 \times 6 = 3$$

$$\text{Solving: } 3v_2 = 9 \rightarrow v_2 = \mathbf{3 \text{ m/s}}, v_1 = \mathbf{0 \text{ m/s}} \text{ (the 2 kg sphere stops).}$$

Check: 2 m/s and 2 m/s is the perfectly plastic case ($e = 0$); -2 and 4 is the elastic case ($e = 1$).

Exam tip: Write restitution as $(v_2 - v_1) = e(u_1 - u_2)$ with one sign convention throughout.

Q5 Answer: 50

Concept: Impulse = area under the F-t curve = change of momentum.

$$\text{Impulse} = \frac{1}{2} \times 0.04 \times 500 = \mathbf{10 \text{ N}\cdot\text{s}}$$

$$10 = 0.2(v - 0) \rightarrow v = \mathbf{50 \text{ m/s}}$$

Average force = $10/0.04 = 250 \text{ N}$, half the peak, as expected for a triangle.

Exam tip: When a graph is given, impulse is always an area. Do not multiply the peak force by the whole time (that gives 100 m/s).

Q6 Answer: (C) 94.9 mm

Concept: Smooth floor $\rightarrow$ all the KE of the block is stored as spring energy at maximum compression.

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \rightarrow x = v\sqrt{m/k}$$

$$x = 3 \times \sqrt{2/2000} = 3 \times 0.03162 = \mathbf{94.9 \text{ mm}}$$

Exam tip: At maximum compression the block is momentarily at rest. With friction, subtract $\mu mg \cdot x$ from the left side.

Q7 Answer: 5.06 (accept 5.02 to 5.10)

Concept: Work-energy theorem: net work done by all forces = change in KE.

$$\text{Work by gravity} = mg \sin 30^\circ \times 4 = 5 \times 9.81 \times 0.5 \times 4 = 98.1 \text{ J}$$

$$\text{Work by friction} = -\mu mg \cos 30^\circ \times 4 = -0.2 \times 5 \times 9.81 \times 0.866 \times 4 = -33.98 \text{ J}$$

$$\frac{1}{2} \times 5 \times v^2 = 98.1 - 33.98 = 64.12 \text{ J} \rightarrow v = \mathbf{5.06 \text{ m/s}}$$

Exam tip: The mass cancels: $v = \sqrt{2gL(\sin \theta - \mu \cos \theta)}$. Useful for a quick check.

Q8 Answer: (C) 6.25 rad/s

Concept: No external torque about the shaft $\rightarrow$ angular momentum is conserved.

$$0.5 \times 10 = (0.5 + 0.3) \omega$$

$$\omega = 5/0.8 = \mathbf{6.25 \text{ rad/s}}$$

KE falls from 25 J to 15.62 J: 37.5 % is lost in the friction between the discs.

Exam tip: This is the rotational version of a plastic collision. Angular momentum is kept, KE is not.

Q9 Answer: 9.905 (accept 9.85 to 9.95)

Concept: At steady speed, tractive force = resistance + component of weight along the slope.
Power = force $\times$ speed.

$$\text{Weight component} = mg \sin \theta = 1000 \times 9.81 \times (1/20) = 490.5 \text{ N}$$

$$\text{Tractive force} = 490.5 + 500 = 990.5 \text{ N}; v = 36 \text{ km/h} = 10 \text{ m/s}$$

$$P = 990.5 \times 10 = 9905 \text{ W} = \mathbf{9.905 \text{ kW}}$$

Exam tip: '1 in 20' means $\sin \theta = 1/20$ (along the slope) unless the question says otherwise.

Q10 Answer: (C) 28.9 N

Concept: Give the mechanism a virtual displacement $\delta\theta$. Reactions at A and C do no work, so only P and Q appear.

$$\text{Coordinates: } y_B = L \sin \theta, x_C = 2L \cos \theta \rightarrow \delta y_B = L \cos \theta \delta\theta, \delta x_C = -2L \sin \theta \delta\theta$$

$$\delta W = -P \delta y_B - Q \delta x_C = (-PL \cos \theta + 2QL \sin \theta) \delta\theta = 0$$

$$Q = (P/2) \cot \theta = 50 \times \cot 60^\circ = \mathbf{28.87 \text{ N}}$$

Exam tip: Virtual work skips the reactions entirely. Choose coordinates from a fixed point (A) so the signs come out naturally.

Q11 Answer: 9.165 (accept 9.10 to 9.22)

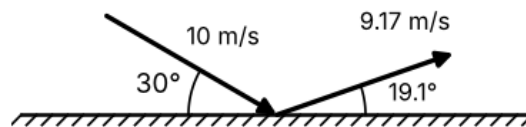
Concept: Smooth floor: the component along the floor is unchanged. Restitution acts only on the normal component.

$$\text{Along the floor: } 10 \cos 30^\circ = \mathbf{8.66 \text{ m/s}} \text{ (unchanged)}$$

$$\text{Normal: } 10 \sin 30^\circ = 5 \text{ m/s} \rightarrow \text{after bounce } 0.6 \times 5 = \mathbf{3 \text{ m/s}}$$

$$v = \sqrt{(8.66^2 + 3^2)} = \mathbf{9.165 \text{ m/s}}, \text{ at } \tan^{-1}(3/8.66) = \mathbf{19.1^\circ} \text{ to the floor}$$

Exam tip: On a smooth surface the ball always leaves at a flatter angle than it arrived (when $e < 1$).

**Q12 Answer: (B) 12 rad/s**

Concept: Linear impulse = change of linear momentum; moment of the impulse about G = change of angular momentum about G.

$$\text{Linear: } 6 = 3 v_G \rightarrow v_G = 2 \text{ m/s}$$

$$\text{Angular: } 6 \times 0.5 = (3 \times 1^2/12) \omega = 0.25 \omega \rightarrow \omega = \mathbf{12 \text{ rad/s}}$$

$$\text{The far end moves at } v_G - \omega(0.5) = 2 - 6 = -4 \text{ m/s, i.e. opposite to the blow.}$$

Exam tip: The point $2/3$ of the length from the struck end is momentarily at rest ($v_G - \omega d = 0 \rightarrow d = 1/6 \text{ m}$ beyond G).

Q13 Answer: 75.0 (accept 74 to 76)

Concept: Energy conservation from release to maximum compression. Gravity does work over the whole drop $h + x$, not just h .

$$mg(h + x) = \frac{1}{2}kx^2 \rightarrow 2 \times 9.81 (0.5 + x) = 2000x^2$$

$$2000x^2 - 19.62x - 9.81 = 0$$

$$x = [19.62 + \sqrt{(19.62^2 + 4 \times 2000 \times 9.81)}] / (2 \times 2000) = \mathbf{75.1 \text{ mm}}$$

Exam tip: Ignoring the extra drop x gives $\sqrt{(2mgh/k)} = 70.0 \text{ mm}$, a typical wrong option.

Q14 Answer: (B) in directions at 90° to each other

Concept: Equal masses, elastic: momentum gives $\vec{u} = \vec{v}_1 + \vec{v}_2$; energy gives $u^2 = v_1^2 + v_2^2$.

$$\text{Squaring the momentum equation: } u^2 = v_1^2 + v_2^2 + 2\vec{v}_1 \cdot \vec{v}_2$$

Comparing: $\vec{v}_1 \cdot \vec{v}_2 = 0 \rightarrow$ the velocities are **perpendicular** (unless the impact is head-on, when the first sphere simply stops).

Exam tip: This is why a snooker cue ball and object ball fly off at right angles after a thin cut.

Q15 Answer: 4.15 (accept 4.10 to 4.20)

Concept: The block does not move until P exceeds limiting friction. Apply impulse-momentum only from that instant.

$$\text{Limiting friction} = 0.2 \times 20 \times 9.81 = 39.24 \text{ N} \rightarrow \text{motion starts at } 10t_0 = 39.24, t_0 = \mathbf{3.924 \text{ s}}$$

$$\text{Impulse of } P \text{ from } t_0 \text{ to } 8 \text{ s} = \int 10t \, dt = 5(8^2 - 3.924^2) = 243.01 \text{ N}\cdot\text{s}$$

$$\text{Impulse of friction} = 39.24 \times (8 - 3.924) = 159.94 \text{ N}\cdot\text{s}$$

$$20v = 243.01 - 159.94 = 83.07 \rightarrow v = \mathbf{4.15 \text{ m/s}}$$

Exam tip: Integrating from $t = 0$ with kinetic friction acting throughout gives only 0.30 m/s, the classic mistake. Before t_0 static friction exactly cancels P .

What next?

Free Day 4 test tonight, 7:00 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with photo, name and marks in the next 8 AM challenge reel.



Website test series



Dig Career Thrust app



Telegram: daily PDFs

Next, Day 5/90: Strength of Materials, Stress, Strain & Mohr's Circle. Comment **GATE90** on our reel to get all links.

Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.